not count on the same 95 percent chance that the 4 percent rule will work for them.

This is a matter where Monte Carlo simulations are able to shine, by allowing simulations to begin from today's starting point rather than incorporating historical outcomes generated from completely different market environments.

Let's consider outcomes for sustainable spending rates using the Monte Carlo simulations with the low interest rate world of today as a starting point for the simulations. This is the simulation approach described in the technical appendix to chapter 3. We will not make any additional adjustments to reflect the high stock market valuation level, but we will preserve the historical equity premium that stocks have earned above bonds. So when bond yields are low, stock returns must be less as well, as there is no particular reason to believe that stocks would offer even higher premiums above bonds than they have done in the past. On average, these simulations allow bond yields (and therefore stock and bond returns) to wander upward toward historical averages over time to reflect the unlikelihood that interest rates would remain low for the next thirty years or longer. But interest rates could remain low, and some simulations account for this; there is just an upward trend on average.

Exhibit 5.6 provides a way to compare the success rates for the 4 percent rule using Monte Carlo simulations that reduce bond yields at the start of retirement. This is compared with simulations calibrated to historical averages, as well as with the results of rolling period historical simulations. This exhibit makes clear that the low interest rate environment creates additional stresses for the 4 percent rule that were not apparent in Monte Carlo simulations calibrated to historical data with higher bond yield assumptions than are available today. For a 50/50 asset allocation to stocks and bonds, these simulations indicate that the 4 percent rule has a 69 percent chance of success instead of a 94 percent chance. The 4 percent rule may work for today's retirees, but it is far from a sure bet or a "safe" spending strategy.